Name: _____

TOPIC TEST 1

Chapter 9: Derivatives

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 6 short answer questions (19 marks)
- Part C: 2 analysis questions (8 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 $\lim_{x \to 4} \frac{x^2 - 16}{3x - 12} =$

A 0

B $\frac{1}{3}$

C $\frac{4}{3}$

D $\frac{8}{3}$

E none of the above

[1 mark]

2 The derivative function of $f(x) = 4x^3$ has the rule

A $f'(x) = 4x^2$

B $f'(x) = 4x^4$

C $f'(x) = 24x$

D $f'(x) = 12x$

E $f'(x) = 12x^2$

[1 mark]

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3 The derivative of $y = 22x^3$ using first principles can be found by.

A $f'(x) = \dfrac{22(x+h)^3 - 22x^3}{h}$

B $f'(x) = \dfrac{(x+h)^3 - x^3}{h}$

C $f'(x) = \lim\limits_{x \to 0}\left(\dfrac{22(x+h)^3 - 22x^3}{h}\right)$

D $f'(x) = \lim\limits_{h \to 0}\left(\dfrac{22(x+h)^3 - 22x^3}{h}\right)$

E $f'(x) = \lim\limits_{h \to 0}\left(\dfrac{22(x+h)^3 - 22x^3}{x}\right)$

[1 mark]

4 The equation of the tangent to the graph of $y = x^2$ at the point (a, a^2) is

A $y = 2a^2x + a$

B $y = 2ax - a^2$

C $y = -2a^2x + a$

D $y = ax$

E $y = 2x + a$

[1 mark]

5 The temperature of a heating steel alloy is given by the equation $T = 2t^2$, where T is the temperature in °C and time is in t seconds. The instantaneous rate of change in temperature at $t = 4$ is:

A 16°C/sec

B 32°C/sec

C 8°C/sec

D 30°C/sec

E 40°C/sec

[1 mark]

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Part B

6 short answer questions
19 marks
Show your working where appropriate.

1 For the function $f(x) = 10x^2 + 3x + 1$

a find $f(2)$

b find $f(2 + h)$

c find the gradient of the secant between the point where $x = 2$ and $x = 2 + h$

d hence, choose appropriate values for h to find $f'(2)$.

[4 marks]

2 $P(x, y)$ and $Q(x + h, y + k)$ are two points on the curve with equation $f(x) = 11x^2$.

a Find the gradient of PQ.

b Hence, find the gradient of the tangent at P.

[3 marks]

3 A model helicopter is flown so that it its height above the ground is given by $h = 3t^2$ at t seconds after it leaves the ground, where the height is in metres. Find each of the following.

a Its height after 3 seconds.

b Its velocity after 3 seconds.

[3 marks]

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4 Let $f(x) = 2x^6$. Find

a $f'(3)$

b x such that $f'(x) = -3$

[3 marks]

5 If the curve $y = ax^2$ has a gradient of 1 at the point (2, 1), find a.

[2 marks]

6 Let $s(t) = \frac{4t^3}{3}$ be the displacement function of a particle moving in a straight line, where t is in seconds and s is in metres. Find the acceleration when the velocity is 1.

[4 marks]

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Part C

2 analysis questions
8 marks
Show your working where appropriate.

1 The cost, in tens of thousands of dollars, of airing x television commercials during an AFL football game is given by $C(x) = 0.6x^2$. Find:

a the exact cost of airing 5 commercials during an AFL football game.

b the average cost of airing the first 5 commercials.

c the marginal cost function and use it to find how fast the cost is increasing when $x = 5$.

[4 marks]

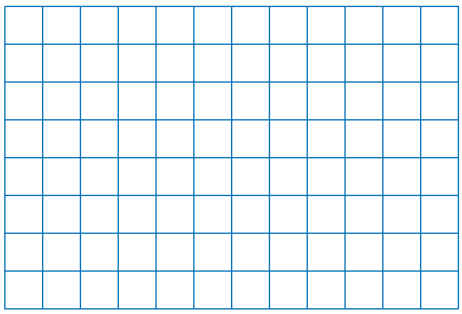
2 A cyclist spends 2 hours cycling from the foothills of the Great Dividing Range to Lake Eppalock. The ride can be described in three stages:

Stage 1: He rides at a constant speed of 20 km/h for 45 minutes.

Stage 2: He rides at a constant speed of 15 km/h for 45 minutes.

Stage 3: He rides at a constant speed of 25 km/h for 30 minutes.

Draw a distance−time graph that illustrates this motion.



[4 marks]
Total marks: 32

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Answers

Part A

[1 mark each]

1 $\lim\limits_{x\to 4}\dfrac{x^2-16}{3x-12}=\lim\limits_{x\to 4}\dfrac{(x-4)(x+4)}{3(x-4)}=\lim\limits_{x\to 4}\dfrac{x+4}{3}=\dfrac{8}{3}$

$\therefore$ **D** [1 mark]

2 Using the rule for finding the derivative of a polynomial:

$f(x) = 4x^3 \Rightarrow f'(x) = 3 \times 4x^{3-1} = 12x^2$

$\therefore$ **E** [1 mark]

3 $f'(x)=\lim\limits_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$

$=\lim\limits_{h\to 0}\dfrac{22(x+h)^3-22x^3}{h}$

$\therefore$ **D** [1 mark]

4 The derivative of the function is $y' = 2x$. At the point (a, a^2), the tangent has gradient $2a$. The equation of the tangent in point slope form is:

$y - a^2 = 2a(x - a)$

$y = 2ax - 2a^2 + a^2$

$= 2ax - a^2$

$\therefore$ **B** [1 mark]

5 The derivative of the function is $\dfrac{dT}{dt}=4t$. Substituting $t = 4$ into this equation, we get $\dfrac{dT}{dt}=16$.

$\therefore$ **A** [1 mark]

Part B

1 a $f(2) = 10(2)^2 + 3(2) + 1 = 47$ [1 mark]

b $f(2 + h) = 10(2 + h)^2 + 3(2 + h) + 1$

$= 10(4 + 4h + h^2) + 6 + 3h + 1$

$= 40 + 40h + 10h^2 + 6 + 3h + 1$

$= 10h^2 + 43h + 47$ [1 mark]

c $\dfrac{f(2+h)-f(2)}{h}=\dfrac{10h^2+43h+47-47}{h}$

$= 10h + 3$ [1 mark]

d $f'(2)=\lim\limits_{h\to 0}(10h+43)$

$= 43$ [1 mark]

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2 $P(x, y)$ and $Q(x + h, y + k)$ are two points on the curve with equation $f(x) = 11x^2$.

a Gradient $PQ = \dfrac{11(x+h)^2 - 11x^2}{h}$ **[1 mark]**

$= \dfrac{11x^2 + 22xh + 11h^2 - 11x^2}{h}$

$= 22x + 11h$ **[1 mark]**

b Gradient of tangent at $P = \lim\limits_{h \to 0}(22x + 11h)$

$= 22x$ **[1 mark]**

3 **a** $h = 3(3)^2$

$= 27$ m **[1 mark]**

b $v = 6t$ **[1 mark]**

$= 6(3)$

$= 18$ m/sec **[1 mark]**

4 **a** $f'(x) = 12(x)^5$

$f'(3) = 12(3)^5 = 2916$ **[1 mark]**

b $f'(x) = 12(x)^5$

$12(x)^5 = -3$ **[1 mark]**

$x^5 = -\dfrac{1}{4}$

$x = \sqrt[5]{-\dfrac{1}{4}}$ **[1 mark]**

5 $\dfrac{dy}{dx} = 2ax$ **[1 mark]**

$1 = 4a$

$\dfrac{1}{4} = a$ **[1 mark]**

6 $v(t) = 4t^2$

$1 = 4t^2$

$\dfrac{1}{4} = t^2$

$\dfrac{1}{2} = t, \quad t > 0$

$a(t) = 8t$

$a\left(\dfrac{1}{2}\right) = 8 \times \dfrac{1}{2}$

$= 4$ m^2/s **[1 mark]**

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Part C

1 a $C(5) = 0.6(5)^2 = \$150\,000$ **[1 mark]**

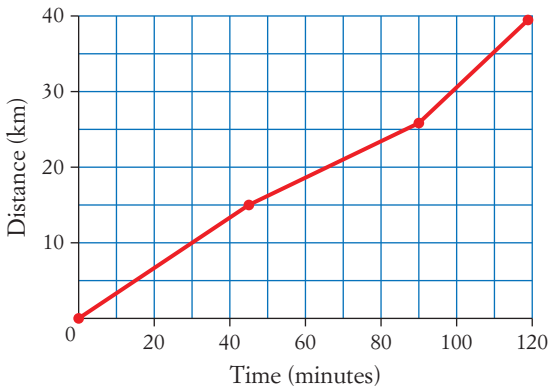
b $\frac{150000}{5} 5\,\$30000$, the average cost of airing the first 5 commercials. **[1 mark]**

c $C'(x) = 1.2x$

$C'(5) = 1.2(5) = 6$

$\therefore$ marginal cost $= \$60\,000$

2



[4 marks]